

Weekly Progress Report

Blender Project

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This report summarises the topics learned and the progress made over the seven weeks of the Blender project.

Week 1

During the first week, I became familiar with the fundamentals of Blender's Geometry Nodes system. I learned how procedural node networks work, generated terrain using noise textures, and explored how different noise parameters influence the resulting landscape.

Week 2

This week focused on enhancing the procedural terrain. I learned height-based masking for assigning materials according to elevation, created materials using the Shader Editor, modelled simple objects from basic geometric primitives, and explored techniques for procedurally distributing objects across the terrain.

Week 3

I continued learning material creation in the Shader Editor and studied the basics of animation in Blender. I also modified the procedural terrain by introducing a river, allowing the landscape to change dynamically through Geometry Nodes.

Week 4

This week involved further object modelling and an introduction to weight painting. I also learned how to apply constraints on object placement so that procedurally generated objects could be restricted to suitable regions of the terrain.

Week 5

I was introduced to Blender's Python API and learned the fundamentals of scripting in Blender. Using Python, I edited exposed inputs of Geometry Node groups to automate and control procedural generation.

Week 6

This week focused on chunk-based terrain generation as a method for efficiently generating larger procedural environments. I also continued learning the Blender Python API and gained further experience with scripting and automation.

Week 7

In the final week, I integrated my knowledge of Geometry Nodes and Python by developing a Blender add-on. The add-on enabled editing of Geometry Node parameters and randomisation of generation seeds, allowing multiple terrain variations to be generated conveniently.